

YASH RANA

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Montreal, QC | **Open to relocation to Toronto** | Eligible to work in Canada (PGWP)

Profile

Master of Applied Computer Science graduate (Concordia University) with **1.5+ years** of applied ML and **Generative AI** experience across industry internships and research. Built and shipped **LLM systems end to end** – retrieval-augmented generation (RAG), multimodal retrieval over video, and abstractive summarization – using **Python, LangChain, FastAPI, Transformers, and vector databases**, with deployment via **Docker** and experiment tracking in **MLflow**. Co-author of a peer-reviewed ML paper; comfortable working in cross-functional teams and translating technical results for non-technical stakeholders.

Education

Concordia University <i>Master of Applied Computer Science (MApCompSc)</i>	Sep 2024 – May 2026 <i>Montreal, QC</i>
Gujarat Technological University <i>Bachelor of Engineering in Computer Engineering</i>	Sep 2020 – Apr 2024 <i>Gujarat, India</i>

Generative AI Projects & Research

Video Scene Intelligence – Multimodal RAG over Video <i>Source Code</i>	LLMs RAG Qdrant FastAPI
- Built a full-stack multimodal RAG system answering natural-language questions about uploaded video with timestamped citations: keyframe extraction (FFmpeg), speech transcription (Whisper), frame captioning (Gemini Vision), and semantic retrieval over a Qdrant vector database - Designed the chunking, embedding, and retrieval strategy across transcript and visual modalities, tuning chunk size and top-k retrieval to improve answer groundedness and reduce hallucination - Architected the async backend (FastAPI, Celery/Redis, PostgreSQL, MinIO) with Docker Compose orchestration and SSE streaming to a Next.js frontend	
Abstractive Text Summarization System <i>Source Code</i>	Transformers NLP Python
Built a modular transformer-based summarization pipeline for long unstructured documents, with separated stages for ingestion, preprocessing, model inference, and evaluation, packaged for reproducible re-runs and deployment	
Early Detection of Diabetes using Machine Learning <i>Published Paper</i>	Python scikit-learn SVM
Peer-reviewed publication (Atlantis Press, ICAA-AI 2025) presenting an SVM-based diagnostic model reaching 95% accuracy, benchmarked via ROC-AUC against Logistic Regression, Random Forest, and KNN	

Experience

Blue Data Consulting <i>Data Science Intern</i>	Jul 2023 – Jun 2024 <i>Surat, India</i>
- Designed and deployed an LLM-based generation system producing structured and unstructured outputs (text, audio, video) using LangChain and cloud AI services, covering the full lifecycle from prototyping to production deployment - Built reproducible preprocessing and prompt-engineering pipelines and tracked experiments with MLflow, cutting end-to-end generation time from 25 to 8 minutes (–68%) while improving output consistency across runs - Collaborated with engineering and business stakeholders in iterative sprint cycles, translating requirements into model and pipeline changes	
GEXEL Telecom International <i>Technical Specialist, Network & Connectivity</i>	Jun 2026 – Present <i>Montreal, QC</i>
- Apply systematic root-cause analysis to diagnostic telemetry (signal strength, throughput, line quality, error rates) to isolate faults across hardware, configuration, and infrastructure layers; identify recurring fault signatures to accelerate triage - Communicate technical diagnoses to non-technical users while operating within PCI-DSS-compliant systems under strict data-handling and access-control protocols	
The Sparks Foundation <i>Data Analyst / Data Science Intern</i>	Feb 2023 – May 2023 <i>Remote</i>
Performed exploratory data analysis, feature engineering, and supervised predictive modeling on tabular datasets using Pandas, NumPy, and scikit-learn, with rigorous data cleaning and quality control; communicated results through weekly dashboards and analytical reports for non-technical stakeholders	

Technical Skills

Generative AI & LLMs: LLMs, Retrieval-Augmented Generation (RAG), LangChain, prompt engineering, Transformers/BERT, Hugging Face, Whisper, Gemini Vision, embeddings & vector search (Qdrant)
Machine Learning & Data: PyTorch, scikit-learn, XGBoost, MLflow, Pandas, NumPy, feature engineering, EDA, statistical modeling, Matplotlib, Seaborn, Power BI
Programming: Python, SQL, Java, JavaScript, TypeScript
Backend & MLOps: FastAPI, Flask, REST APIs, Celery/Redis, Docker & Docker Compose, Git, GitHub Actions, CI/CD, Linux
Cloud & Data Stores: AWS, Azure, GCP (Gemini API); PostgreSQL, MongoDB, MinIO, Supabase
Spoken Languages: English (fluent), French (professional working proficiency)